

# Joint AF + PRF — parameter diagnostics

*fits-dir = /data/ds-retsupp/derivatives/af\_prf\_joint\_full*

total fits : 240  
unique subjects : 30  
modes : signed

fits per ROI:

V1 : 30  
V2 : 30  
V3 : 30  
V3AB : 30  
hV4 : 30  
L0 : 30  
T0 : 30  
V0 : 30

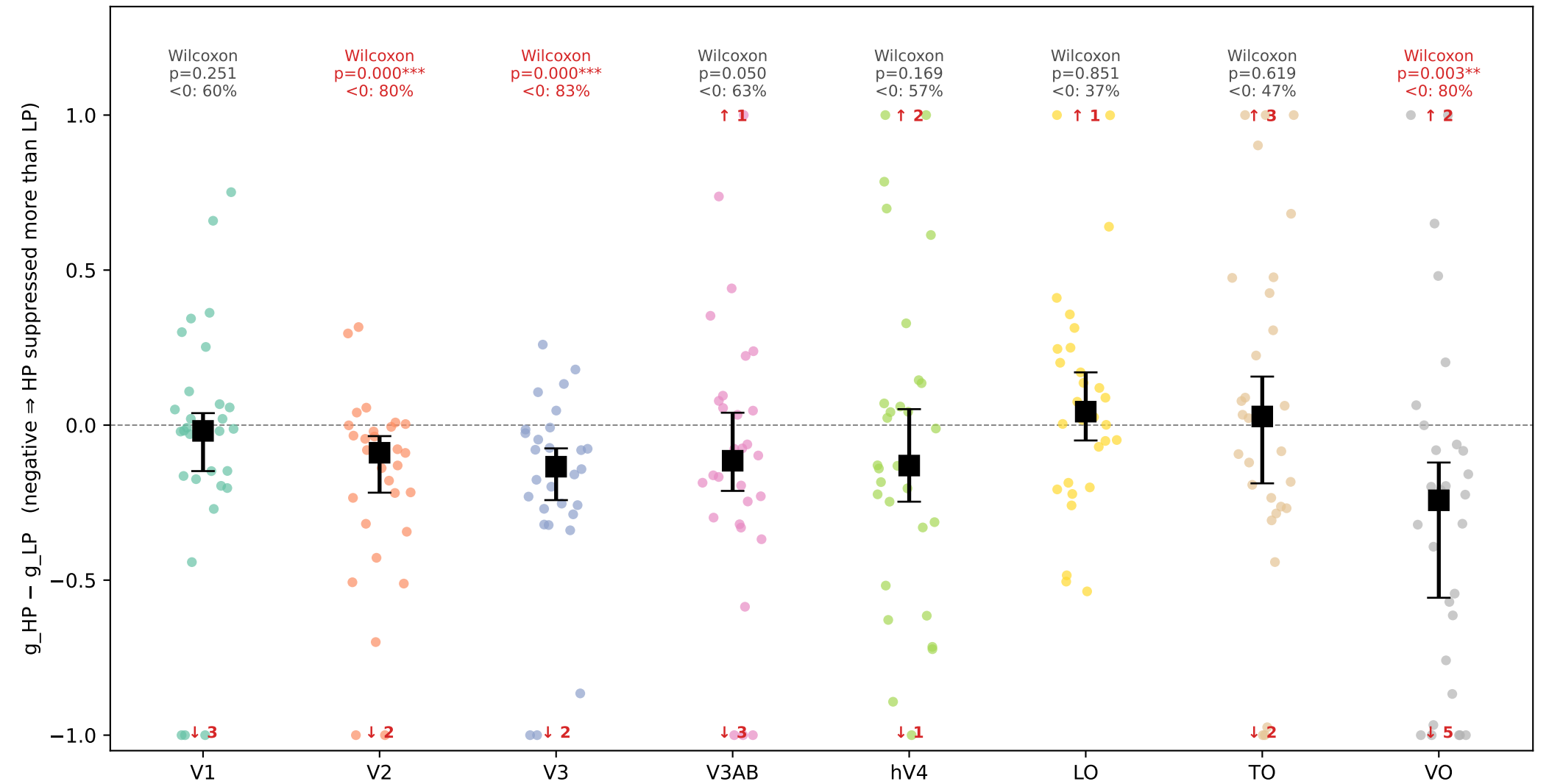
Diagnostics in this report:

- Per-ROI distributions of  $\sigma_{AF}$ ,  $g_{HP}$ ,  $g_{LP}$ ,  $g_{HP} - g_{LP}$ ,  $R^2$ .
- Pairwise scatter + Pearson  $r$  of the 3 shared AF params.
- Per-fit dot plot (one row per (subject, ROI)).
- $g_{HP}$  vs  $g_{LP}$  coloured by ROI — diagonal = no HP differential.

### Per-ROI summary (median across fits)

| roi  | n_fits | sigma_AF_med | g_HP_med | g_LP_med | g_diff_med | r2_mean |
|------|--------|--------------|----------|----------|------------|---------|
| V1   | 30     | 2.093        | 0.260    | 0.309    | -0.019     | 0.132   |
| V2   | 30     | 2.292        | 0.076    | 0.158    | -0.089     | 0.139   |
| V3   | 30     | 2.360        | 0.096    | 0.232    | -0.134     | 0.125   |
| V3AB | 30     | 2.465        | 0.238    | 0.229    | -0.115     | 0.040   |
| hV4  | 30     | 2.040        | 0.316    | 0.241    | -0.131     | 0.061   |
| L0   | 30     | 2.782        | 0.162    | 0.069    | 0.043      | 0.094   |
| T0   | 30     | 2.567        | 0.153    | 0.048    | 0.028      | 0.043   |
| V0   | 30     | 2.336        | 0.615    | 0.722    | -0.242     | 0.026   |

HP-vs-LP test ( $g_{HP} < g_{LP}$ ): per-fit median  $\pm$  95 % bootstrap CI (outlier-resistant; 27/240 fits with  $|g_{HP} - g_{LP}| > 1.0$  clipped, count shown red  $\uparrow/\downarrow$ )

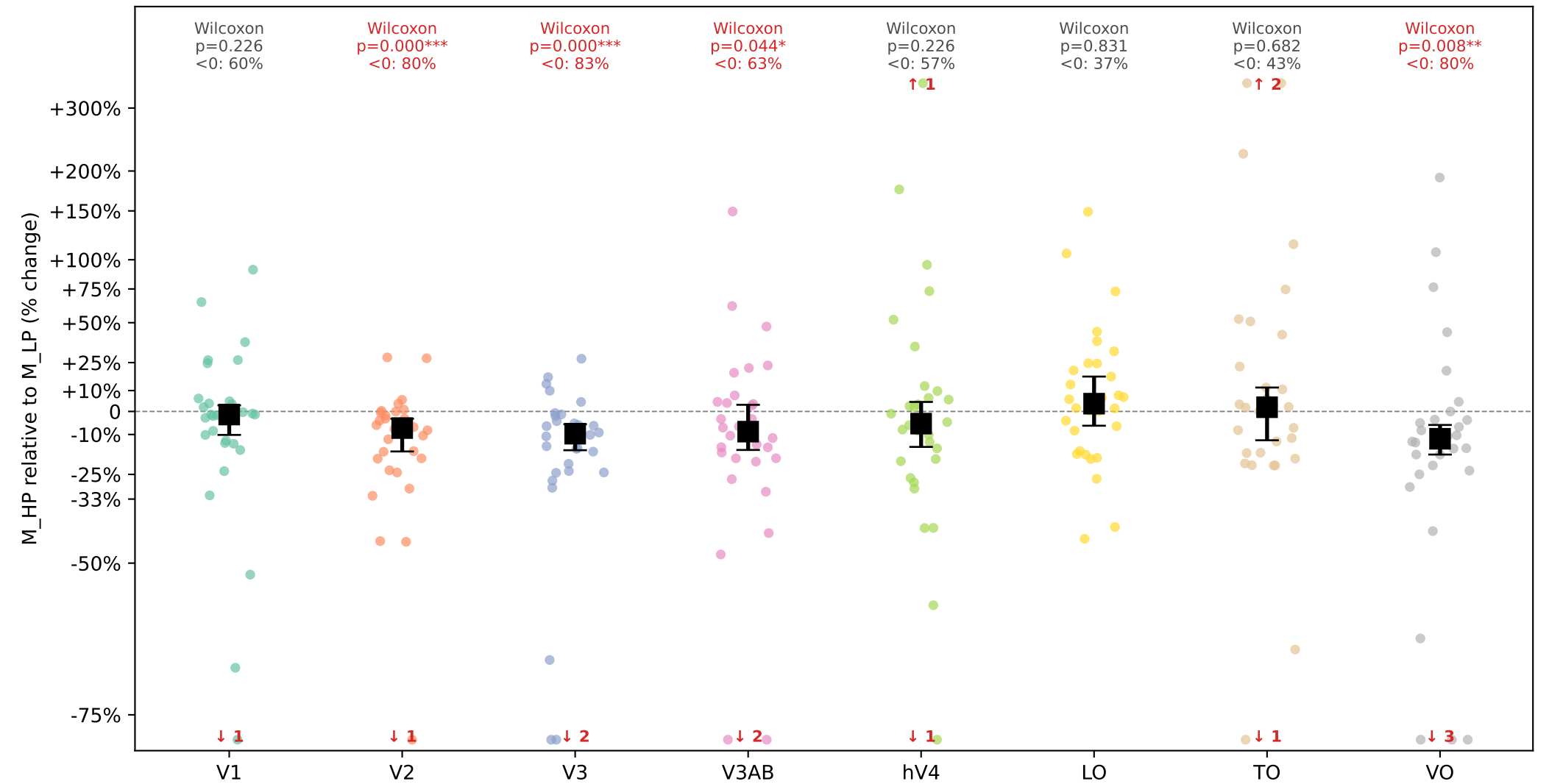


# Per-ROI HP-vs-LP test — $H_0: g_{HP} - g_{LP} \geq 0$

Alternative:  $g_{HP} < g_{LP}$  (HP-specific suppression)

| ROI  | n  | $g_{HP} - g_{LP}$ mean $\pm$ SEM | t      | p (t, $1t$ 0) | W       | p (Wilcoxon, $1t$ 0) | frac $g_{HP} - g_{LP} < 0$ |
|------|----|----------------------------------|--------|---------------|---------|----------------------|----------------------------|
| V1   | 30 | -0.264 $\pm$ 0.202               | -1.309 | 0.100         | 199.000 | 0.251                | 0.600                      |
| V2   | 30 | -0.397 $\pm$ 0.231               | -1.715 | 0.048         | 67.000  | 0.000                | 0.800                      |
| V3   | 30 | -0.210 $\pm$ 0.066               | -3.182 | 0.002         | 67.000  | 0.000                | 0.833                      |
| V3AB | 30 | -0.384 $\pm$ 0.233               | -1.646 | 0.055         | 152.000 | 0.050                | 0.633                      |
| hV4  | 30 | +0.027 $\pm$ 0.159               | 0.172  | 0.568         | 185.000 | 0.169                | 0.567                      |
| L0   | 30 | +0.083 $\pm$ 0.069               | 1.201  | 0.880         | 283.000 | 0.851                | 0.367                      |
| T0   | 30 | +0.137 $\pm$ 0.147               | 0.930  | 0.820         | 247.000 | 0.619                | 0.467                      |
| V0   | 30 | -0.490 $\pm$ 0.195               | -2.507 | 0.009         | 99.000  | 0.003                | 0.800                      |

5-HP test ( $M_{HP} < M_{LP}$  (scale-invariant)): per-fit median  $\pm$  95 % bootstrap CI (outlier-resistant; 14/240 fits with  $|\log((1+g_{HP})/(1+g_{LP}))| > 1.5$  clipped, count

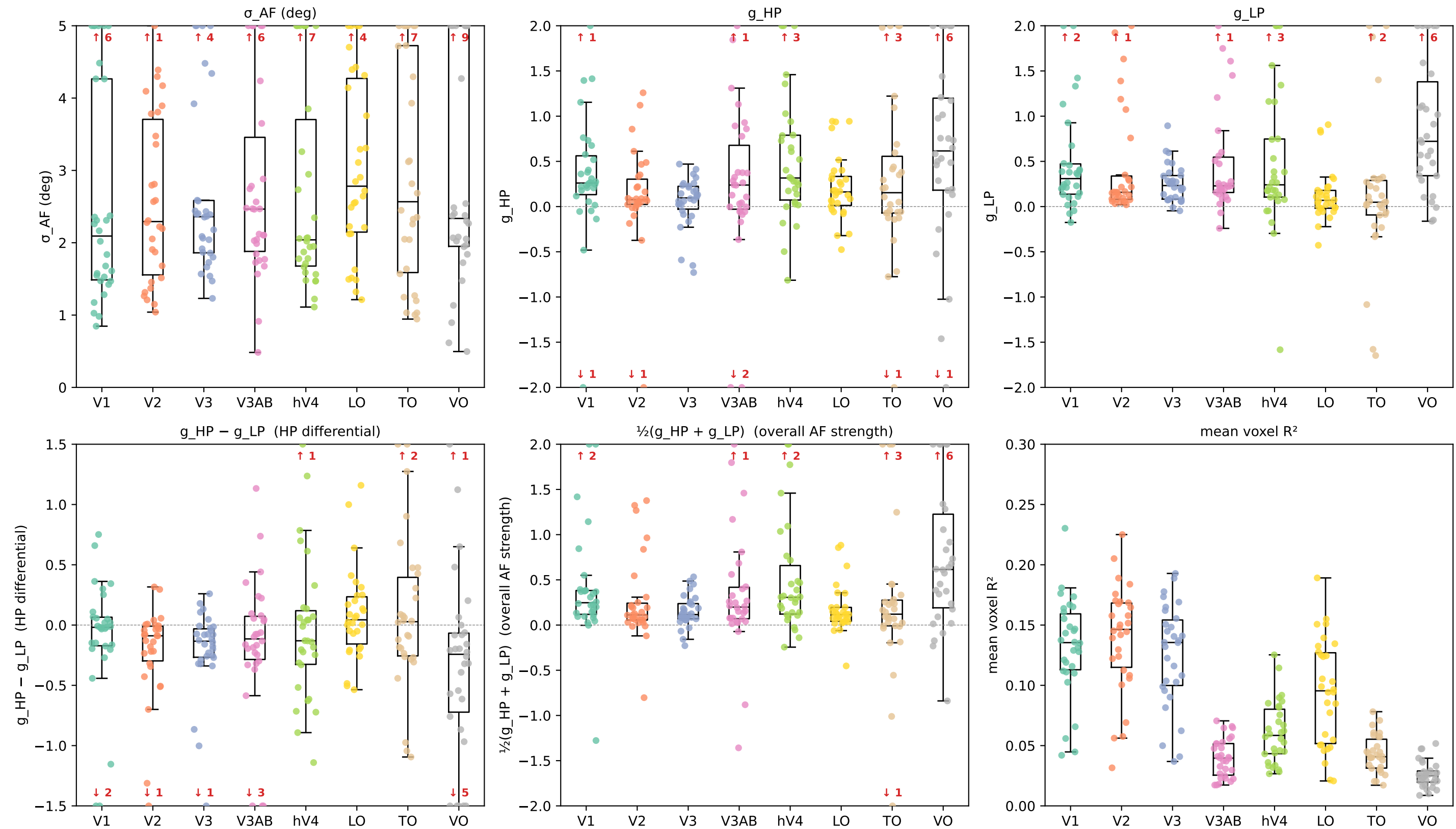


# Per-ROI HP-vs-LP test — H0: $\log((1+g_{HP})/(1+g_{LP})) \geq 0$

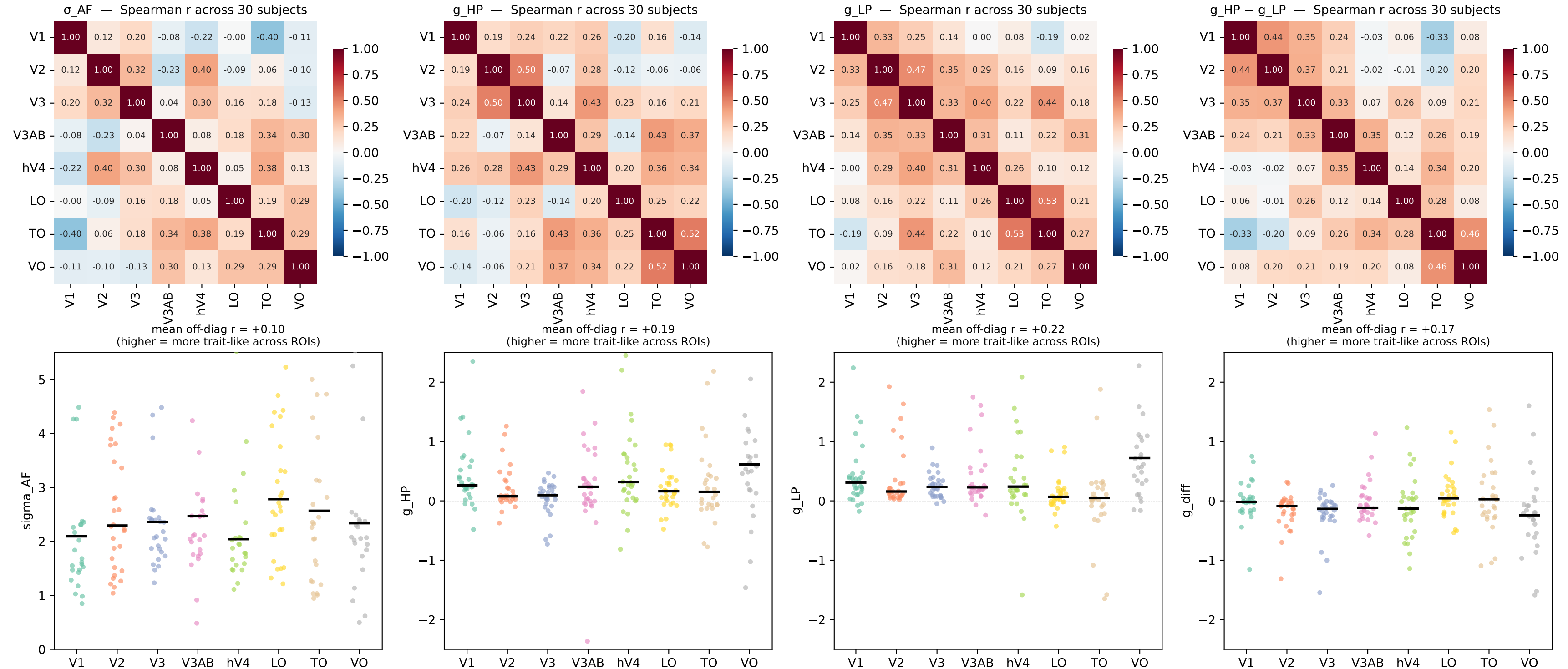
Alternative:  $M_{HP} < M_{LP}$  (scale-invariant) (HP-specific suppression)

| ROI  | n  | $\log((1+g_{HP})/(1+g_{LP}))$ | mean $\pm$ SEM     | t      | p (t, $\text{lt } 0$ ) | W       | p (Wilcoxon, $\text{lt } 0$ ) | frac $\log((1+g_{HP})/(1+g_{LP}))$ |
|------|----|-------------------------------|--------------------|--------|------------------------|---------|-------------------------------|------------------------------------|
| V1   | 30 |                               | -0.166 $\pm$ 0.140 | -1.179 | 0.124                  | 195.000 | 0.226                         |                                    |
| V2   | 30 |                               | -0.257 $\pm$ 0.143 | -1.789 | 0.042                  | 69.000  | 0.000                         |                                    |
| V3   | 30 |                               | -0.229 $\pm$ 0.081 | -2.810 | 0.004                  | 68.000  | 0.000                         |                                    |
| V3AB | 30 |                               | -0.281 $\pm$ 0.177 | -1.583 | 0.062                  | 149.000 | 0.044                         |                                    |
| hV4  | 30 |                               | +0.049 $\pm$ 0.170 | 0.286  | 0.612                  | 195.000 | 0.226                         |                                    |
| L0   | 30 |                               | +0.058 $\pm$ 0.059 | 0.983  | 0.833                  | 279.000 | 0.831                         |                                    |
| T0   | 30 |                               | +0.208 $\pm$ 0.181 | 1.148  | 0.870                  | 255.000 | 0.682                         |                                    |
| V0   | 30 |                               | -0.410 $\pm$ 0.211 | -1.945 | 0.031                  | 117.000 | 0.008                         |                                    |

Per-ROI parameter distributions across (subject) fits (y-axis clipped; red ↑/↓ = #fits past the limit)



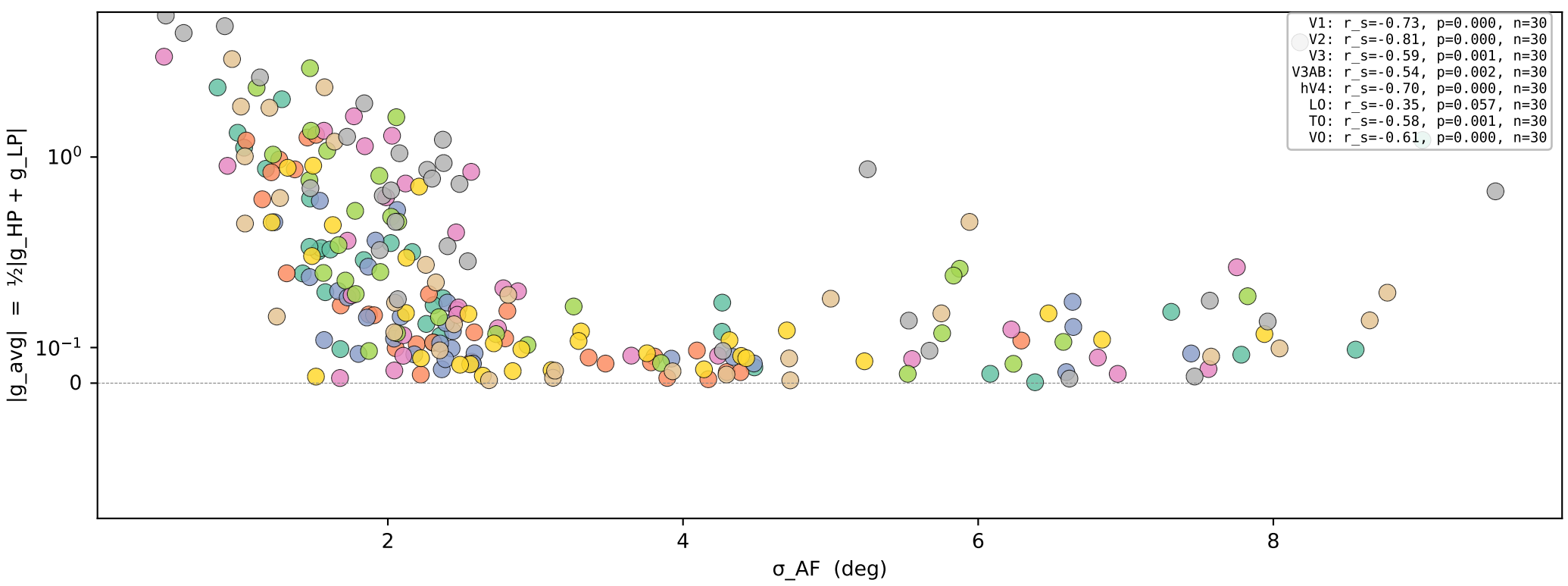
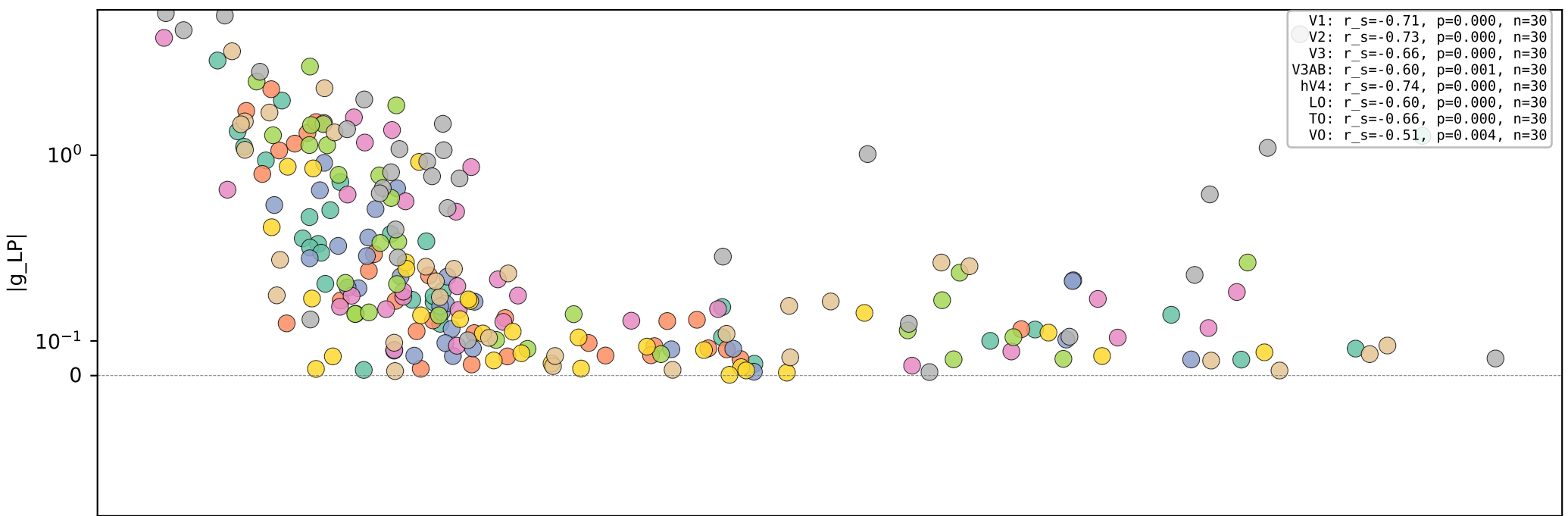
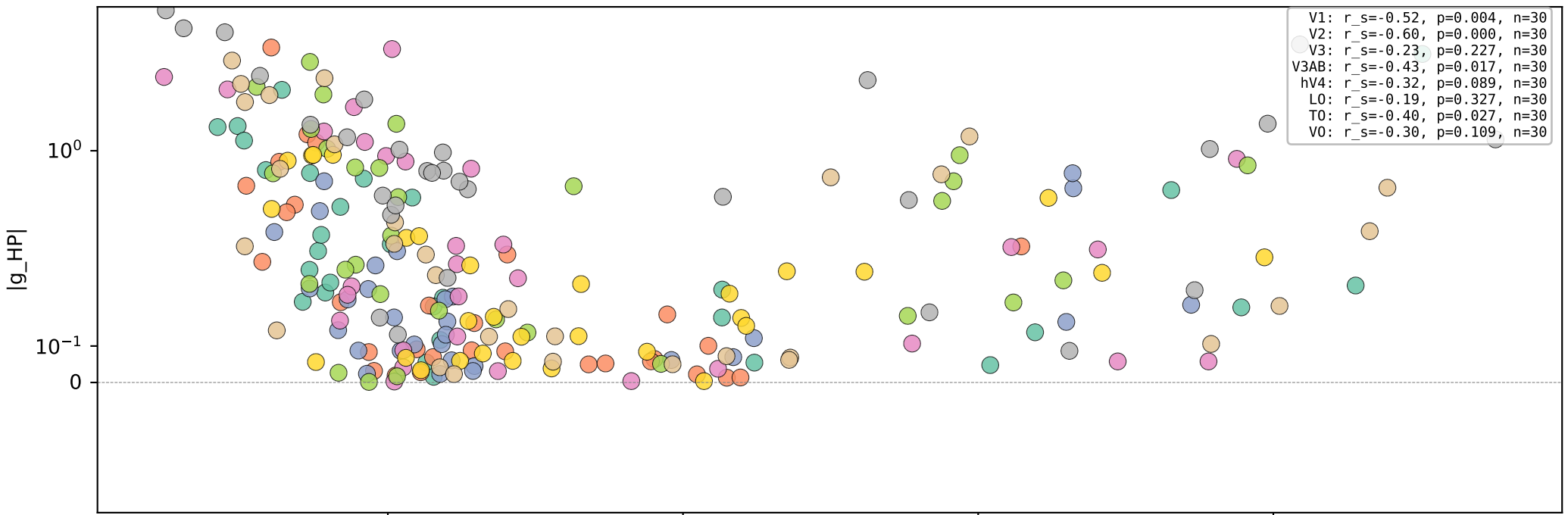
Inter-ROI stability of AF parameters across subjects.  
Are these traits of the subject (high inter-ROI r), or specific to each visual area (low r)?

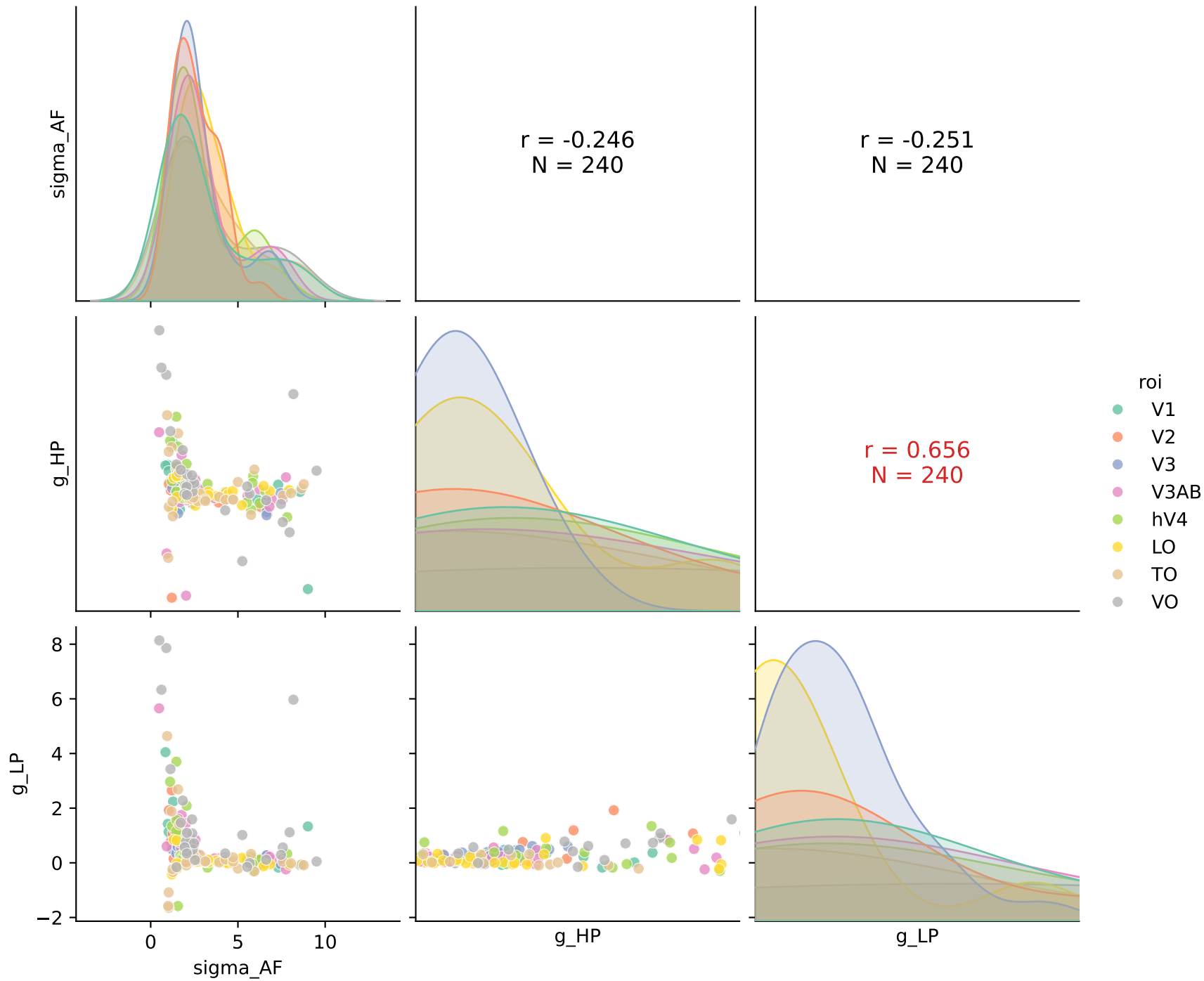




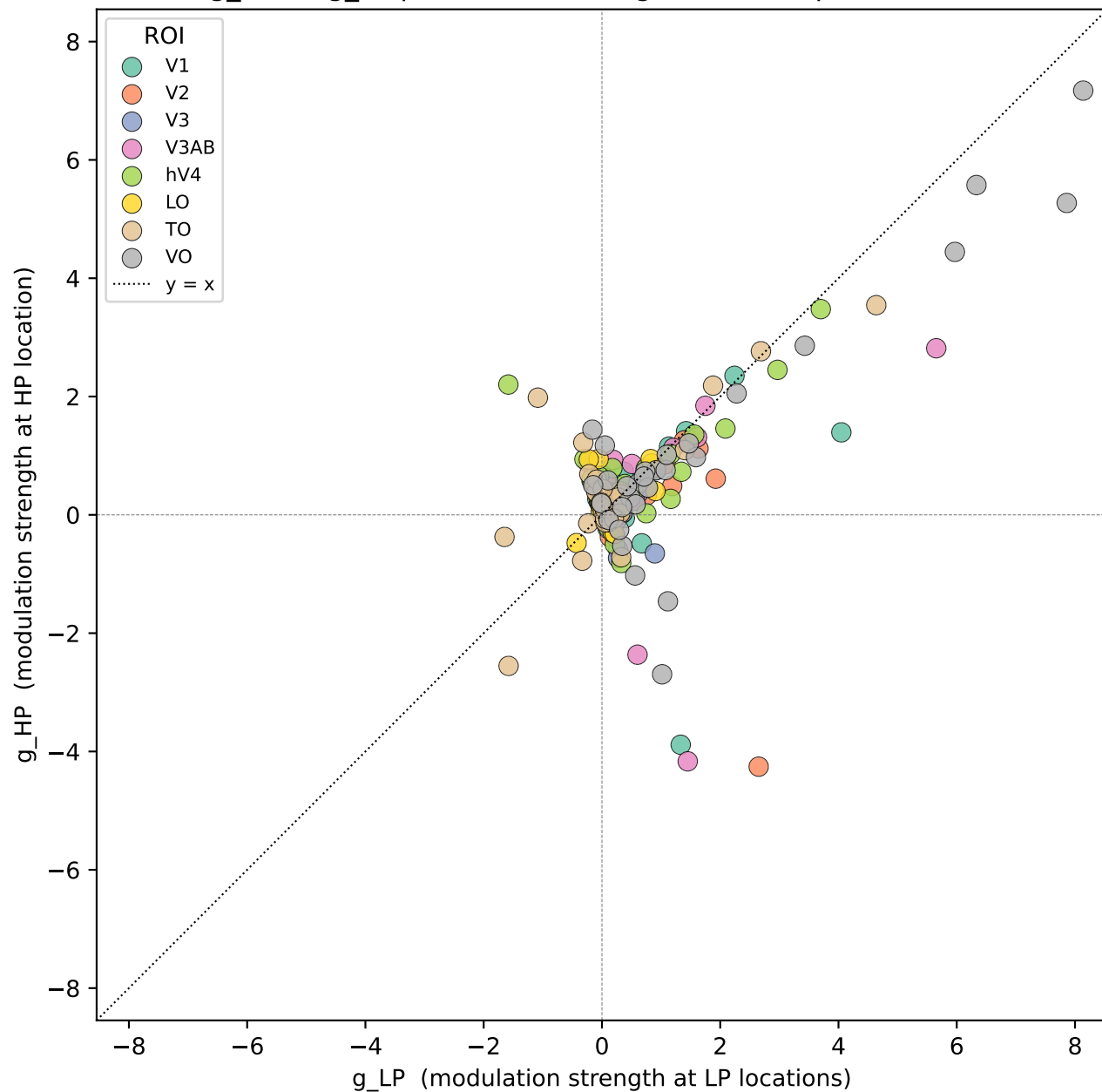
● V1    ● V3    ● hV4    ● TO  
● V2    ● V3AB    ● LO    ● V0

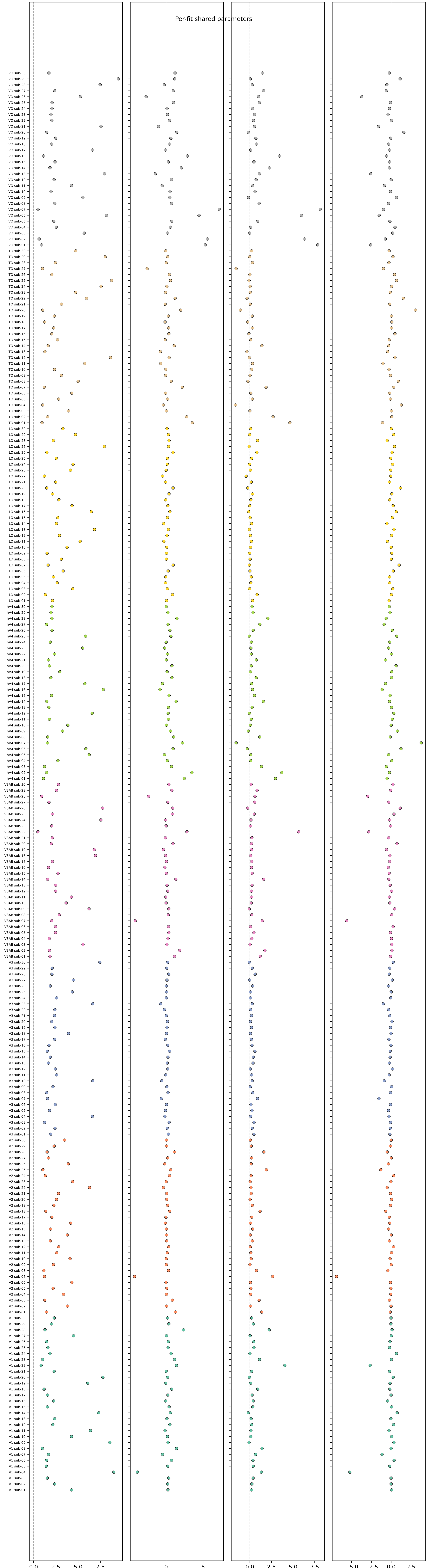
Identifiability check:  $\sigma_{AF}$  vs  $|gain|$  per ROI across subjects.  
 Strong  $\sigma_{AF} \leftrightarrow |gain|$  correlation = trade-off; flat clouds = ROI-coherent fit.





g\_HP vs g\_LP per fit — off-diagonal = HP-specific effect





## Predicted vs observed shifts — what each side is

PER VOXEL × HP-CONDITION row, we have:

base PRF center  $(x_v, y_v)$   
= the voxel's position parameters from the JOINT AF + Gaussian-PRF fit to the BOLD time-series (the same fit that produced  $\sigma_{AF}$ ,  $g_{HP}$ ,  $g_{LP}$ ). NOT the standard mean-PRF fit.

predicted AD-PRF center  $(\mu_{v_C})$   
= closed-form center-of-mass of  $p_{v_C}(g) = M_C(g) \cdot S_v(g)$  computed on a discretized 2D grid, where  $M_C(g)$  is built from the SHARED-AF parameters ( $\sigma_{AF}$ ,  $g_{HP}$ ,  $g_{LP}$ , mode) for the condition C, and  $S_v(g)$  is the voxel's Gaussian PRF. Negative modulation is clamped to 0 before the COM (matching the braincoder forward model).

observed conditionwise PRF center  $(x^{obs}_{v_C}, y^{obs}_{v_C})$   
= the  $(x, y)$  parameters from `prf_conditionfit/model4` — INDEPENDENT per-condition Gaussian-PRF fits, one per HP block, run on subsets of runs. These are NOT used by the joint AF + PRF fit.

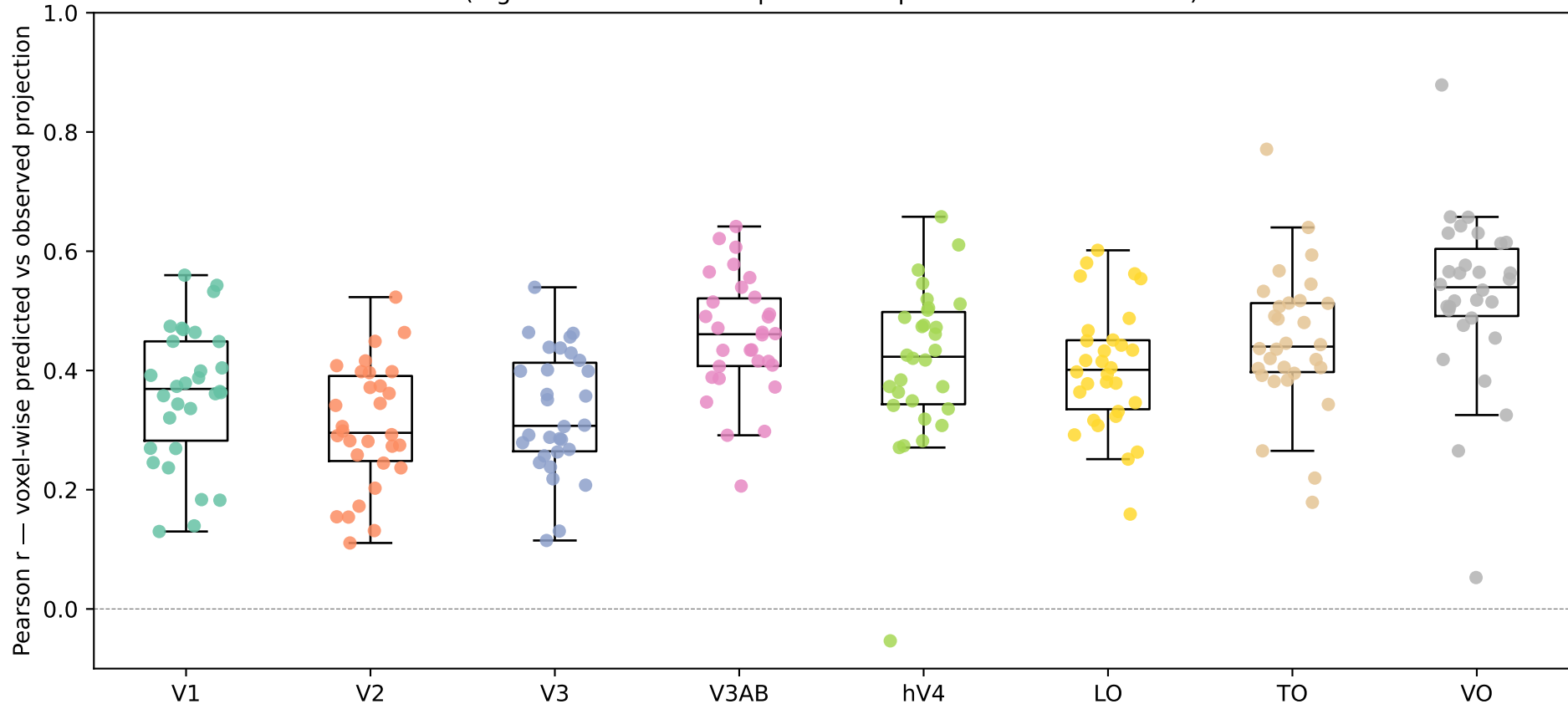
The two correlation axes:

proj (1D) : project (predicted – base) and (observed – base) shifts onto the unit vector pointing AWAY from the HP location of that condition C. Negative = toward HP, positive = away from HP.

$(\Delta x, \Delta y)$  (2D): full per-component shift vectors  
 $\Delta x = x_{pred} - base_x$ ;  $\Delta y = y_{pred} - base_y$   
(and similarly for observed).

High Pearson  $r$  between predicted and observed at the voxel level means the joint BOLD-level AF model genuinely recovers the voxel-wise shift pattern that the conditionwise PRF fits independently report.

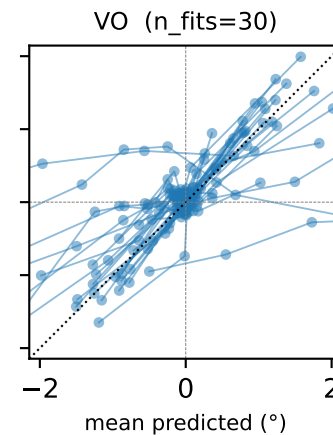
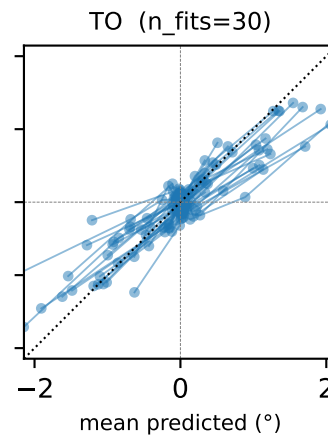
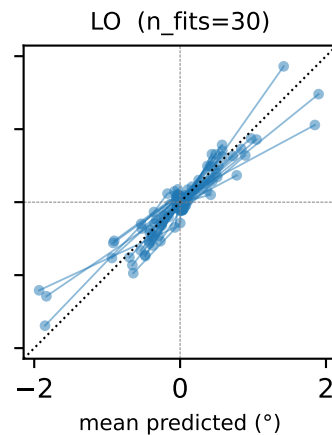
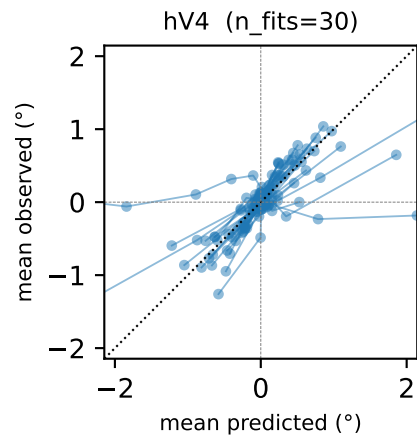
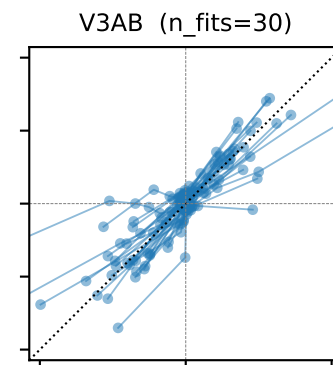
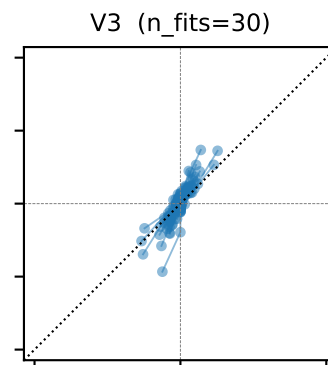
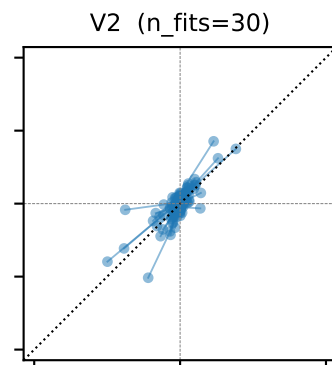
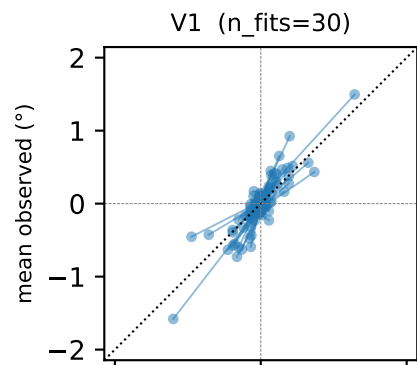
Per-fit predicted-vs-observed shift correlation  
(higher  $r$  = AF model captures real per-voxel shift structure)



### Per-ROI Pearson r (predicted vs observed)

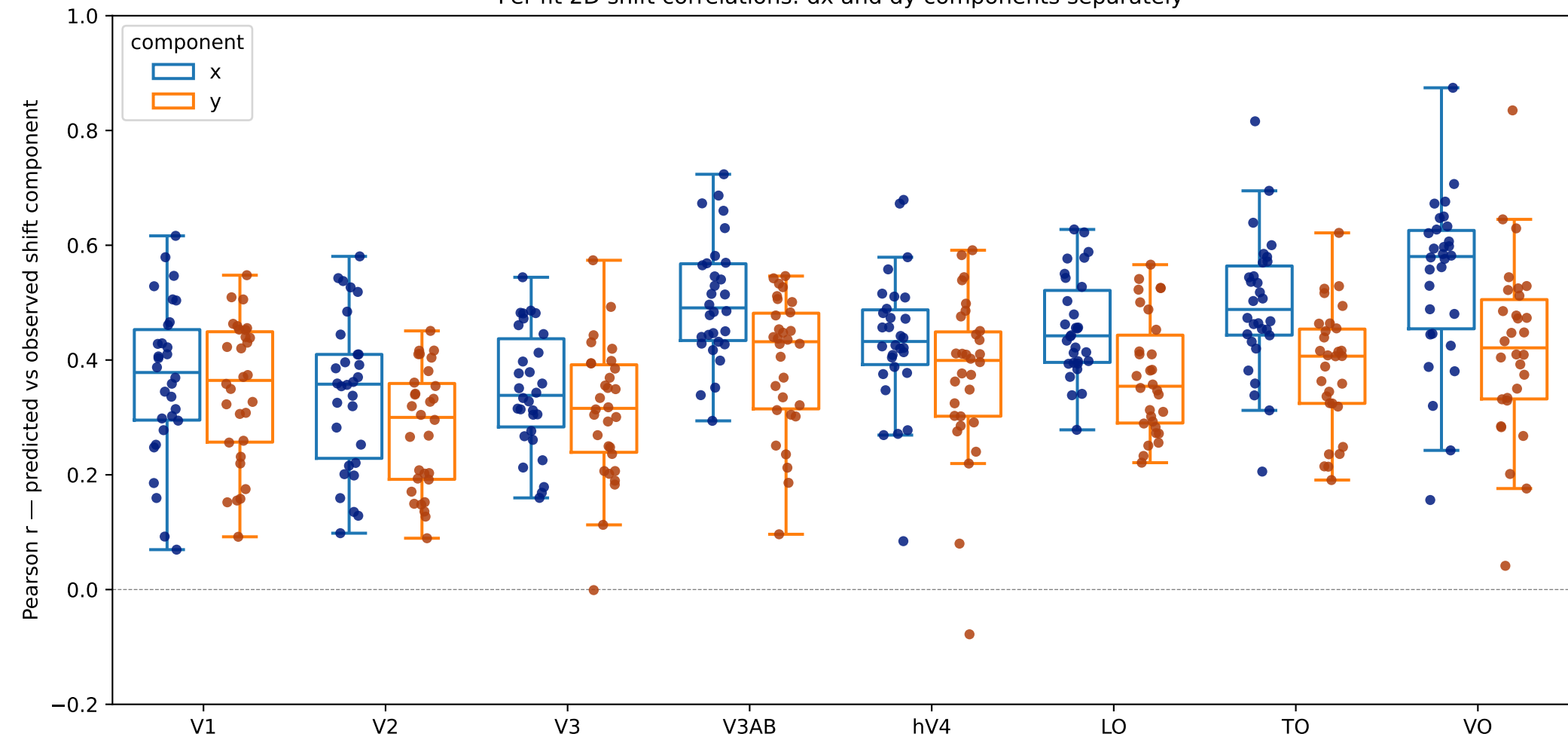
| roi  | count | mean  | median | min    | max   |
|------|-------|-------|--------|--------|-------|
| V1   | 30    | 0.362 | 0.369  | 0.130  | 0.560 |
| V2   | 30    | 0.307 | 0.295  | 0.111  | 0.523 |
| V3   | 30    | 0.330 | 0.307  | 0.115  | 0.540 |
| V3AB | 30    | 0.457 | 0.461  | 0.206  | 0.642 |
| hV4  | 30    | 0.414 | 0.423  | -0.053 | 0.658 |
| L0   | 30    | 0.405 | 0.401  | 0.159  | 0.602 |
| T0   | 30    | 0.451 | 0.440  | 0.179  | 0.771 |
| V0   | 30    | 0.524 | 0.540  | 0.053  | 0.879 |

Calibration: bin-averaged predicted vs observed (one line per fit)  
On-diagonal = AF prediction matches observed shift on average

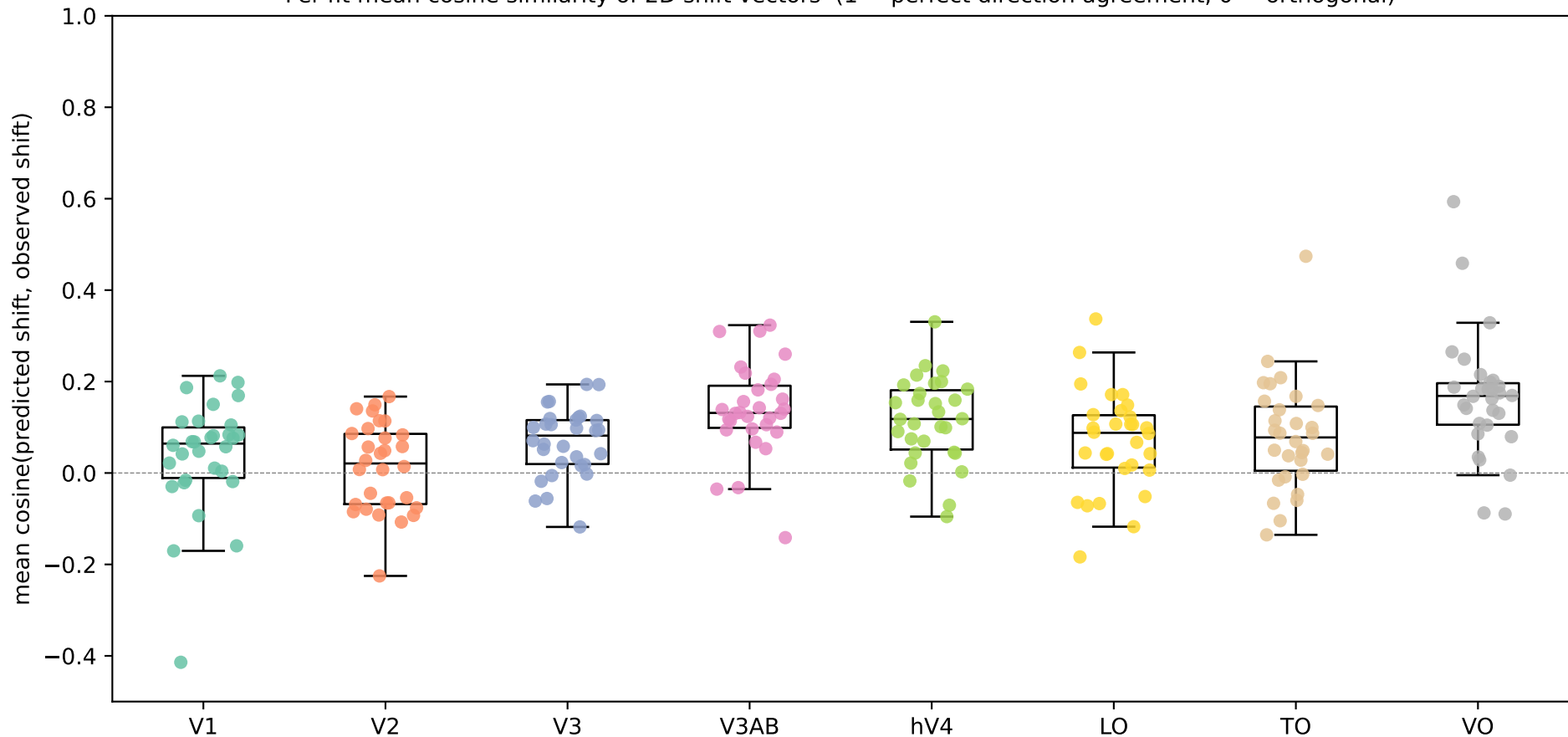




Per-fit 2D shift correlations: dx and dy components separately



Per-fit mean cosine similarity of 2D shift vectors (1 = perfect direction agreement, 0 = orthogonal)



### Per-ROI 2D shift agreement (medians across fits)

| roi  | n_fits | r_x_med | r_y_med | cos_med |
|------|--------|---------|---------|---------|
| V1   | 30     | 0.378   | 0.364   | 0.064   |
| V2   | 30     | 0.358   | 0.300   | 0.021   |
| V3   | 30     | 0.338   | 0.316   | 0.082   |
| V3AB | 30     | 0.491   | 0.432   | 0.131   |
| hV4  | 30     | 0.432   | 0.399   | 0.118   |
| L0   | 30     | 0.442   | 0.354   | 0.088   |
| T0   | 30     | 0.488   | 0.407   | 0.078   |
| V0   | 30     | 0.580   | 0.421   | 0.169   |

Pooled per-voxel 2D shift density (N = 618,148 voxel-condition rows; hexbin, log scale)

